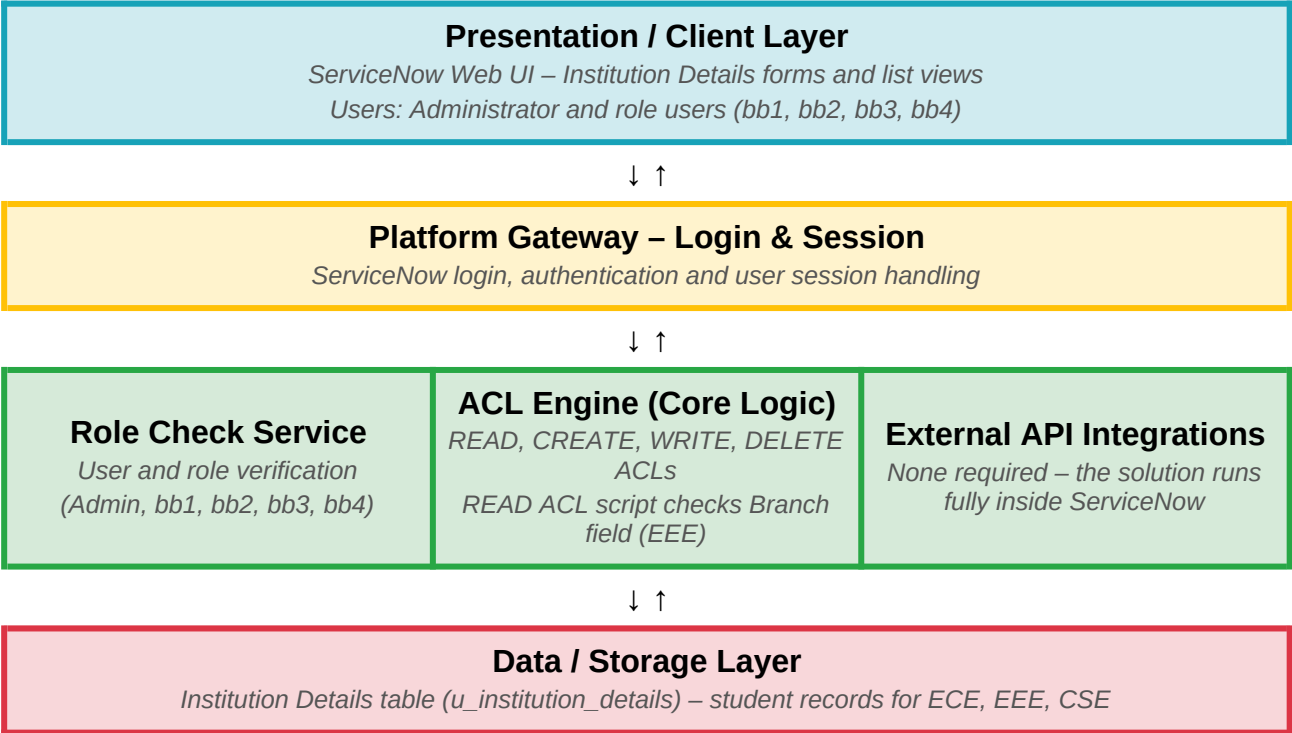


Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-1374
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture of the Script-Controlled ACL solution: User → Role → ACL → Institution Details table → Student records. The user's assigned role is checked by the ACL before access to a record is allowed.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Forms and list views where the administrator and role users view, create, edit and delete Institution Details records.	ServiceNow Web UI (Forms & Lists)
Platform Gateway	Handles user login and sessions and identifies who is making each request.	ServiceNow authentication and sessions
Access Control Services	Checks the user's role and applies the READ, CREATE, WRITE and DELETE ACLs. The READ ACL script also checks the Branch value to allow EEE records only for the authorized role. Admin has full access.	ServiceNow Roles (bb1–bb4), Access Control (ACL), server-side JavaScript
Database	Stores student information in the custom Institution Details table.	ServiceNow database (u_institution_details)

Table Design – Institution Details (u_institution_details)

Field	Type
Student Roll Number	Auto Number
Student Name	Reference – User
Faculty Name	Reference – User
Branch	Choice (ECE, EEE, CSE)
Email	String
Phone Number	String
Description	Multi String

ACL Design

ACL	Operation	Required Role
READ ACL	Read	bb1
CREATE ACL	Create	bb2
WRITE ACL	Write	bb3
DELETE ACL	Delete	bb4

The READ ACL additionally checks the Branch field and permits access to EEE records for the authorized role.